

MA4110/MA5020: Computational Methods for Fluid Flow

Tutorial Sheet 2

Finite Difference Approximations, Truncation Error, Von Neumann Stability, and Modified Equations

Instructor: Biswarup Biswas

Question 1: Derivation of Finite Difference Stencils by Undetermined Coefficients

- (a) **Second-Order One-Sided Forward Difference:** Near a physical inlet boundary at $x = x_0$, centered differences cannot be evaluated because nodes to the left do not exist in the computational domain.

- i. Using the three-point forward stencil $\{u_j, u_{j+1}, u_{j+2}\}$, express the approximation as:

$$u'(x_j) \approx \frac{a u_j + b u_{j+1} + c u_{j+2}}{\Delta x}$$

Expand $u_{j+1} = u(x_j + \Delta x)$ and $u_{j+2} = u(x_j + 2\Delta x)$ in Taylor series up to $\mathcal{O}(\Delta x^3)$, determine the undetermined coefficients a, b, c to achieve the highest possible order of accuracy, and show that:

$$u'(x_j) = \frac{-3u_j + 4u_{j+1} - u_{j+2}}{2\Delta x} + \frac{\Delta x^2}{3} u'''(x_j) + \mathcal{O}(\Delta x^3)$$

- ii. What is the formal order of accuracy of this boundary stencil?
- (b) **Second Derivative on a Non-Uniform Mesh:** In boundary layers, grid points are often clustered non-uniformly. Consider a non-uniform grid with unequal spacing: $h_1 = x_j - x_{j-1}$ and $h_2 = x_{j+1} - x_j$.
- i. Formulate the three-point approximation for the second derivative:

$$u''(x_j) \approx c_{-1} u_{j-1} + c_0 u_j + c_1 u_{j+1}$$

By Taylor-expanding u_{j-1} and u_{j+1} about x_j , find the coefficients c_{-1}, c_0, c_1 that eliminate the $u(x_j)$ and $u'(x_j)$ terms while producing $u''(x_j)$ with unit coefficient.

- ii. Verify that when $h_1 = h_2 = \Delta x$, your formula reduces to the standard central difference $\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$.
- iii. What is the leading truncation error term when $h_1 \neq h_2$? What happens to the order of accuracy if the ratio $h_2/h_1 \neq 1$ as $h_1, h_2 \rightarrow 0$?
- (c) **High-Order Central Difference for the Laplacian:** Using a symmetric 5-point stencil $\{u_{j-2}, u_{j-1}, u_j, u_{j+1}, u_{j+2}\}$, derive the fourth-order accurate central difference formula for the second derivative:

$$u''(x_j) = \frac{-u_{j+2} + 16u_{j+1} - 30u_j + 16u_{j-1} - u_{j-2}}{12\Delta x^2} + \mathcal{O}(\Delta x^4)$$

Question 2: Local Truncation Error and Consistency

- (a) **Forward-Time Central-Space (FTCS) for the Heat Equation:** Consider the one-dimensional heat/diffusion equation $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$ discretized using the explicit FTCS scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \alpha \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

- i. Derive the local truncation error τ_j^n by substituting exact smooth solutions into the difference scheme. Express τ_j^n purely in terms of spatial derivatives using the governing PDE $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$.
 - ii. Show that for an arbitrary choice of grid ratio, the scheme is $\mathcal{O}(\Delta t, \Delta x^2)$.
 - iii. Show that if we choose the special diffusion ratio $r = \frac{\alpha \Delta t}{\Delta x^2} = \frac{1}{6}$, the spatial error terms cancel up to fourth order, yielding an extraordinary accuracy of $\mathcal{O}(\Delta t^2, \Delta x^4)$!
- (b) **The DuFort-Frankel Scheme and Conditional Consistency:** To avoid the severe time-step restriction of FTCS, DuFort and Frankel (1953) modified the unstable Richardson Leapfrog diffusion stencil by replacing u_j^n on the right-hand side with the temporal average $\frac{1}{2}(u_j^{n+1} + u_j^{n-1})$:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = \alpha \left[\frac{u_{j+1}^n - (u_j^{n+1} + u_j^{n-1}) + u_{j-1}^n}{\Delta x^2} \right]$$

- i. By expanding all terms in Taylor series about (x_j, t^n) , show that the local truncation error satisfies:

$$\tau_j^n = \alpha \left(\frac{\Delta t}{\Delta x} \right)^2 \frac{\partial^2 u}{\partial t^2} + \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^2)$$

- ii. Explain why the DuFort-Frankel scheme is **conditionally consistent**: show that it is consistent with the parabolic heat equation $u_t = \alpha u_{xx}$ only if $\frac{\Delta t}{\Delta x} \rightarrow 0$ as $\Delta t, \Delta x \rightarrow 0$.
- iii. What continuous partial differential equation does the scheme *actually* solve if Δt and Δx vanish at a fixed ratio $\frac{\Delta t}{\Delta x} = c$? What is the mathematical type (hyperbolic, parabolic, elliptic) of that resulting equation?

Question 3: Von Neumann Stability Analysis for Hyperbolic Systems

Recall that for linear difference schemes with constant coefficients, the round-off error mode $\epsilon_j^n = \hat{\epsilon}^n e^{ij\xi}$ satisfies the exact same difference equation as the numerical solution u_j^n . Stability requires that the amplification factor $G(\xi) \equiv \frac{\epsilon_j^{n+1}}{\epsilon_j^n}$ satisfy $|G(\xi)| \leq 1$ for all phase angles $\xi = k\Delta x \in [-\pi, \pi]$. Let $\nu = \frac{a\Delta t}{\Delta x}$ denote the Courant-Friedrichs-Lewy (CFL) number.

- (a) **First-Order Upwind (FOU):** For the advection equation $u_t + au_x = 0$ with $a > 0$, the upwind scheme is:

$$u_j^{n+1} = u_j^n - \nu(u_j^n - u_{j-1}^n)$$

- i. Show that the amplification factor is $G(\xi) = (1 - \nu) + \nu e^{-i\xi}$.
 - ii. Compute $|G(\xi)|^2$ in terms of $\cos \xi$ and show that $|G(\xi)|^2 = 1 - 2\nu(1 - \nu)(1 - \cos \xi)$.
 - iii. Prove that $|G(\xi)| \leq 1$ for all $\xi \in [-\pi, \pi]$ if and only if $0 \leq \nu \leq 1$.
 - iv. What happens to the error mode with the highest frequency ($\xi = \pi$, Nyquist mode) when $\nu = 0.5$ versus when $\nu = 1.0$?
- (b) **Downwind Differencing (The Wrong Upwind):** Suppose an engineer carelessly uses a forward difference for the spatial derivative when $a > 0$:

$$u_j^{n+1} = u_j^n - \nu(u_{j+1}^n - u_j^n)$$

Derive its amplification factor $G_{\text{down}}(\xi)$ and show that $|G_{\text{down}}(\xi)|^2 = 1 + 2\nu(1 + \nu)(1 - \cos \xi)$. Deduce that this downwind scheme is **unconditionally unstable** for any $\nu > 0$.

- (c) **The Leapfrog (CTCS) Scheme:** Consider the second-order central-in-time, central-in-space (Leapfrog) scheme:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

- i. Substitute $u_j^n = G^n e^{ij\xi}$ and derive the quadratic characteristic equation for G :

$$G^2 + 2i\nu \sin \xi G - 1 = 0$$

- ii. Solve this quadratic equation to find the two roots $G_1(\xi)$ and $G_2(\xi)$.
 iii. Show that if $|\nu \sin \xi| \leq 1$, the discriminant is negative, and both roots satisfy $|G_1| = |G_2| = 1$ identically!
 iv. What does $|G| = 1$ for all resolvable frequencies imply about the numerical dissipation of the Leapfrog scheme? What happens if $|\nu| > 1$?

Question 4: Von Neumann Stability for Parabolic Equations: Explicit vs. Implicit

Consider the heat equation $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$ with $\alpha > 0$. Define the diffusion grid ratio $r = \frac{\alpha \Delta t}{\Delta x^2}$.

- (a) **Explicit FTCS Scheme:** For the scheme $u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$:

- i. Show that $G(\xi) = 1 - 4r \sin^2\left(\frac{\xi}{2}\right)$.
 ii. Prove that the stability condition $-1 \leq G(\xi) \leq 1$ requires:

$$r = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2}$$

- iii. If we halve the grid spacing ($\Delta x \rightarrow \Delta x/2$), by what factor must the time step Δt be reduced to maintain stability? Explain why this quadratic scaling poses a severe computational bottleneck for fine meshes.
 (b) **Backward-Time Central-Space (BTCS) Fully Implicit Scheme:** To remove the restrictive time-step barrier, we evaluate the spatial derivatives at the future time level t^{n+1} :

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \alpha \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}$$

- i. Show that the amplification factor is:

$$G(\xi) = \frac{1}{1 + 4r \sin^2\left(\frac{\xi}{2}\right)}$$

- ii. Prove that $|G(\xi)| < 1$ for all non-zero phase angles ξ for **any** $r > 0$, meaning that BTCS is **unconditionally stable**.
 (c) **Crank-Nicolson Scheme:** The Crank-Nicolson method averages the spatial derivative between levels n and $n+1$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\alpha}{2} \left[\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} + \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \right]$$

- i. Derive the amplification factor:

$$G(\xi) = \frac{1 - 2r \sin^2\left(\frac{\xi}{2}\right)}{1 + 2r \sin^2\left(\frac{\xi}{2}\right)}$$

- ii. Prove that Crank-Nicolson is unconditionally stable ($|G(\xi)| \leq 1$ for all $r > 0$).
 iii. Examine the behavior of the highest-frequency mode ($\xi = \pi$) as $r \rightarrow \infty$. Show that $G(\pi) \rightarrow -1$. What undesirable numerical artifacts occur in practice if Crank-Nicolson is run with very large time steps $r \gg 1$?

Question 5: Modified Equation Analysis: Numerical Dissipation and Dispersion

Recall that the modified equation reveals the exact continuous PDE solved by a discrete difference algorithm. Higher-order time derivatives are systematically replaced by spatial derivatives using the governing differential equation $\frac{\partial u}{\partial t} = -a \frac{\partial u}{\partial x}$.

- (a) **Lax-Friedrichs Modified Equation:** Starting from the Lax-Friedrichs difference equation for $u_t + au_x = 0$:

$$\frac{u_j^{n+1} - \frac{1}{2}(u_{j+1}^n + u_{j-1}^n)}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

- i. Show that this scheme can be rewritten as the FTCS scheme plus an artificial viscosity term:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = \frac{\Delta x^2}{2\Delta t} \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \right)$$

- ii. Carry out Taylor expansions and eliminate time derivatives to prove that the modified equation up to second order is:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \frac{\Delta x^2}{2\Delta t} (1 - \nu^2) \frac{\partial^2 u}{\partial x^2}$$

- iii. Compare this artificial diffusion coefficient $\alpha_{\text{LxF}} = \frac{\Delta x^2}{2\Delta t} (1 - \nu^2)$ with the Upwind diffusion coefficient $\alpha_{\text{Upwind}} = \frac{a\Delta x}{2} (1 - \nu)$. Show that for small CFL numbers ($\nu \ll 1$), $\alpha_{\text{LxF}} \rightarrow \infty$ as $\Delta t \rightarrow 0$ with fixed Δx ! Explain why Lax-Friedrichs produces massive smearing of flow features.

- (b) **Second-Order Lax-Wendroff Modified Equation:** The Lax-Wendroff scheme was constructed to eliminate the leading diffusion term $\frac{\partial^2 u}{\partial x^2}$.

- i. By expanding $u_j^{n+1}, u_{j\pm 1}^n$ up to third order and using $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^3 u}{\partial t^3} = -a^3 \frac{\partial^3 u}{\partial x^3}$, derive the modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = -\frac{a\Delta x^2}{6} (1 - \nu^2) \frac{\partial^3 u}{\partial x^3} + \mathcal{O}(\Delta t^3, \Delta x^3)$$

- ii. Identify the leading order error term and explain why Lax-Wendroff does not dissipate wave amplitudes to second order, but instead causes dispersive wave distortion.

- (c) **The Magic CFL Number ($\nu = 1$):**

- i. Substitute $\nu = 1$ into the modified equations of Upwind, Lax-Friedrichs, and Lax-Wendroff. What happens to all numerical error coefficients?
- ii. Evaluate the discrete algebraic update formulas of Upwind and Lax-Wendroff at $\nu = 1$ and show that both simplify to $u_j^{n+1} = u_{j-1}^n$.
- iii. Provide the physical explanation for this exactness using the method of characteristics.

Question 6: Physical Interpretation of Dispersion: Trailing vs. Leading Wiggles

- (a) **Dispersion Relation of Dispersive Modified Equations:** Consider a general modified advection equation containing a leading third-order spatial derivative:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = -C \frac{\partial^3 u}{\partial x^3}$$

where C is a constant determined by the numerical scheme.

- i. Substitute a harmonic Fourier plane wave $u(x, t) = e^{i(kx - \omega t)}$ into this equation.
- ii. Show that the dispersion relation relating frequency ω to wavenumber k is:

$$\omega(k) = ak - Ck^3$$

- iii. The phase velocity of a wave component with wavenumber k is $c(k) = \frac{\omega(k)}{k}$. Deduce that:

$$c(k) = a - Ck^2$$

- (b) **Lax-Wendroff: Trailing (Post-Shock) Oscillations:** From the Lax-Wendroff modified equation, $C = \frac{a\Delta x^2}{6}(1 - \nu^2)$.
- For any stable CFL number $0 < \nu < 1$, verify that $C > 0$.
 - Show that short waves (large wavenumber k) travel at a phase speed $c(k) < a$.
 - Explain why this velocity deficit causes spurious numerical oscillations to lag **behind** (to the left of) the advancing shock wave (as observed in Lecture 8 simulations).
- (c) **Beam-Warming: Leading (Pre-Shock) Oscillations:** The second-order upwind Beam-Warming scheme has modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = + \frac{a\Delta x^2}{6}(2 - 3\nu + \nu^2) \frac{\partial^3 u}{\partial x^3}$$

- For $0 < \nu < 1$, show that $(2 - 3\nu + \nu^2) = (1 - \nu)(2 + \nu) > 0$, so that the third derivative has a **positive** coefficient ($C < 0$).
- Show that for Beam-Warming, short waves travel at a phase speed $c(k) > a$.
- Deduce that high-frequency ripples outrun the wave front, appearing as **leading** (pre-shock) oscillations ahead of the shock.

Question 7: Godunov's Theorem and the Need for Nonlinear Limiters

- (a) **Monotonicity-Preserving Schemes:** A numerical scheme is defined to be **monotonicity preserving** if it does not create new local extrema: that is, if the initial data is monotonically increasing ($u_j^0 \leq u_{j+1}^0$ for all j), the solution remains monotonically increasing for all future time levels ($u_j^n \leq u_{j+1}^n$).

Consider a general linear difference scheme with constant coefficients:

$$u_j^{n+1} = \sum_{k=-p}^q c_k u_{j+k}^n$$

Prove that this linear scheme is monotonicity preserving if and only if all coefficients are non-negative:

$$c_k \geq 0 \quad \text{for all } k \in \{-p, \dots, q\}$$

- (b) **First-Order Upwind is Monotonic:** Write the First-Order Upwind scheme ($a > 0$) in the form $u_j^{n+1} = c_{-1}u_{j-1}^n + c_0u_j^n$. Identify c_{-1} and c_0 , and confirm that both are non-negative provided the CFL condition $0 \leq \nu \leq 1$ holds.
- (c) **Second-Order Lax-Wendroff Violates Monotonicity:**
- Express the Lax-Wendroff update $u_j^{n+1} = c_{-1}u_{j-1}^n + c_0u_j^n + c_1u_{j+1}^n$ by finding the coefficients c_{-1}, c_0, c_1 in terms of ν .
 - Show that:
$$c_1 = -\frac{\nu(1-\nu)}{2}$$
 - Prove that for any stable non-trivial CFL number $0 < \nu < 1$, the coefficient c_1 is **strictly negative** ($c_1 < 0$).
 - Explain what this negative coefficient implies about monotonicity: show that if a step initial profile ($u_j^0 = 1$ for $j \leq 0$ and $u_j^0 = 0$ for $j \geq 1$) is updated by Lax-Wendroff, an overshoot $u_0^1 > 1$ or undershoot $u_1^1 < 0$ is guaranteed to occur!

- (d) **Godunov's Barrier Theorem:** State Godunov's Barrier Theorem (1959). In light of your findings in parts (b) and (c), explain why modern high-resolution CFD methods must use **non-linear adaptive switching** (such as TVD flux limiters) even when solving linear PDEs.